Texture streaming in auditory scenes

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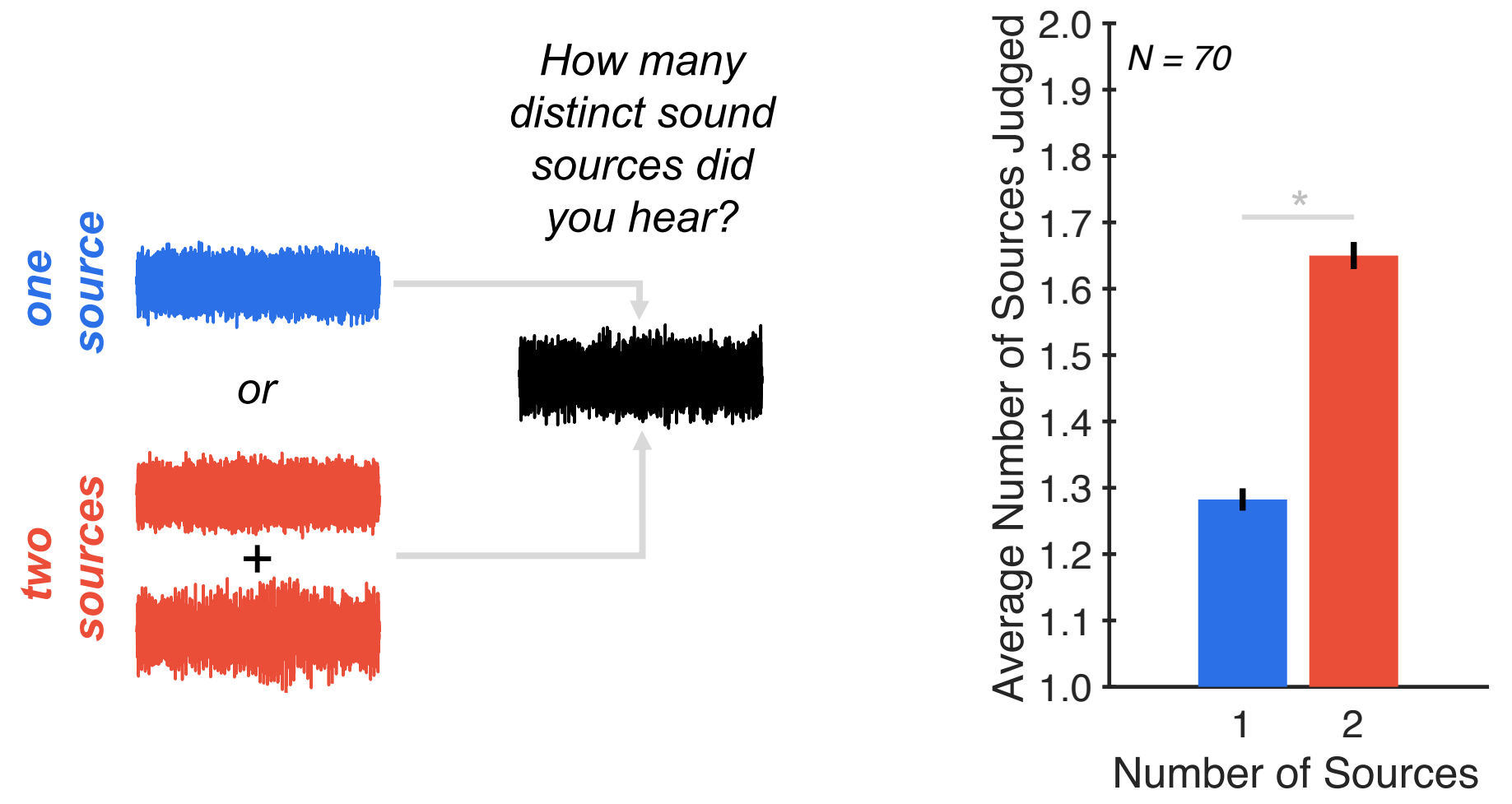
Introduction

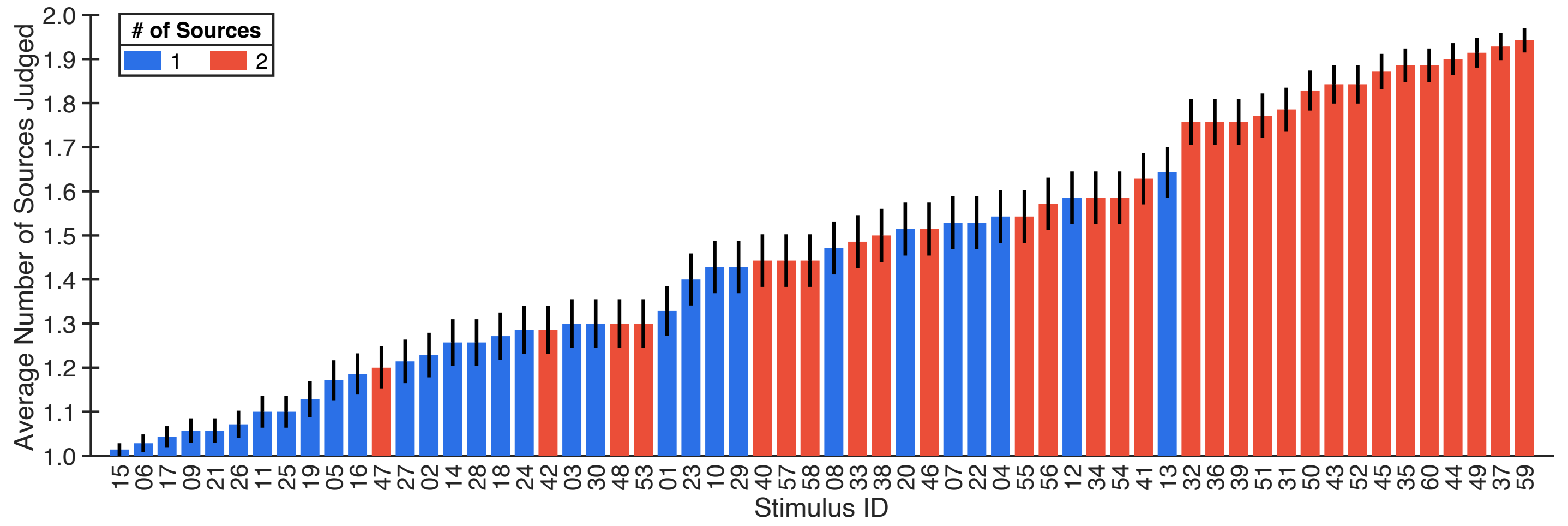
- Classical auditory scene analysis research has primarily been inspired by thinking about speech and music but explored perceptual organization using simple synthetic stimuli (tones and noise)
- Unclear whether classical auditory scene analysis principles apply to realistic auditory scenes with natural sounds
- We explored auditory scene analysis with a common class of real-world environmental sounds—sound textures

Sound textures

- Often result from the superposition of many similar acoustic events (*applause, room of people talking, etc.*)
- Broad and diverse class of environmental sounds
- Ubiquitous in the world, frequently forming the background "noise" in auditory scenes (*traffic noise, air conditioner, rain, etc.*)
- Perceptual representation captured by signal-computable models

Can listeners judge the number of textures in an auditory scene?

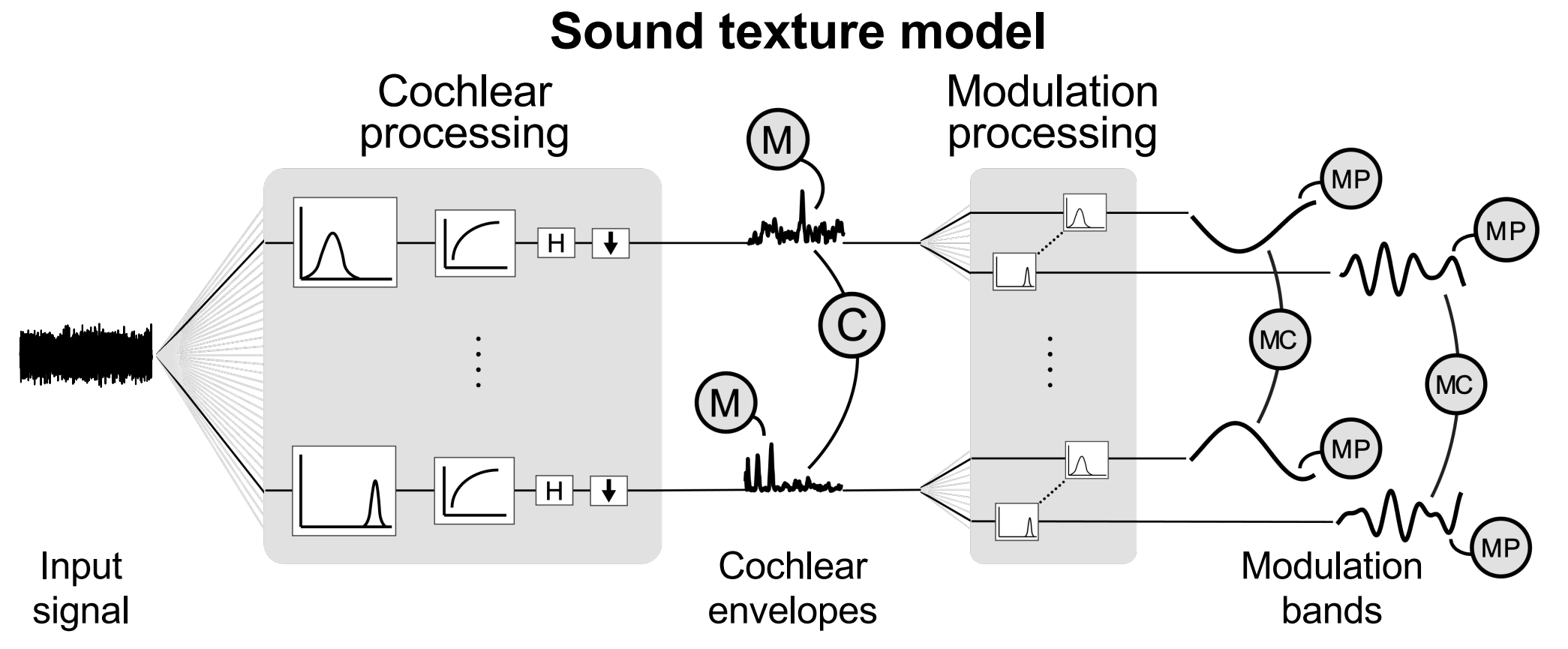




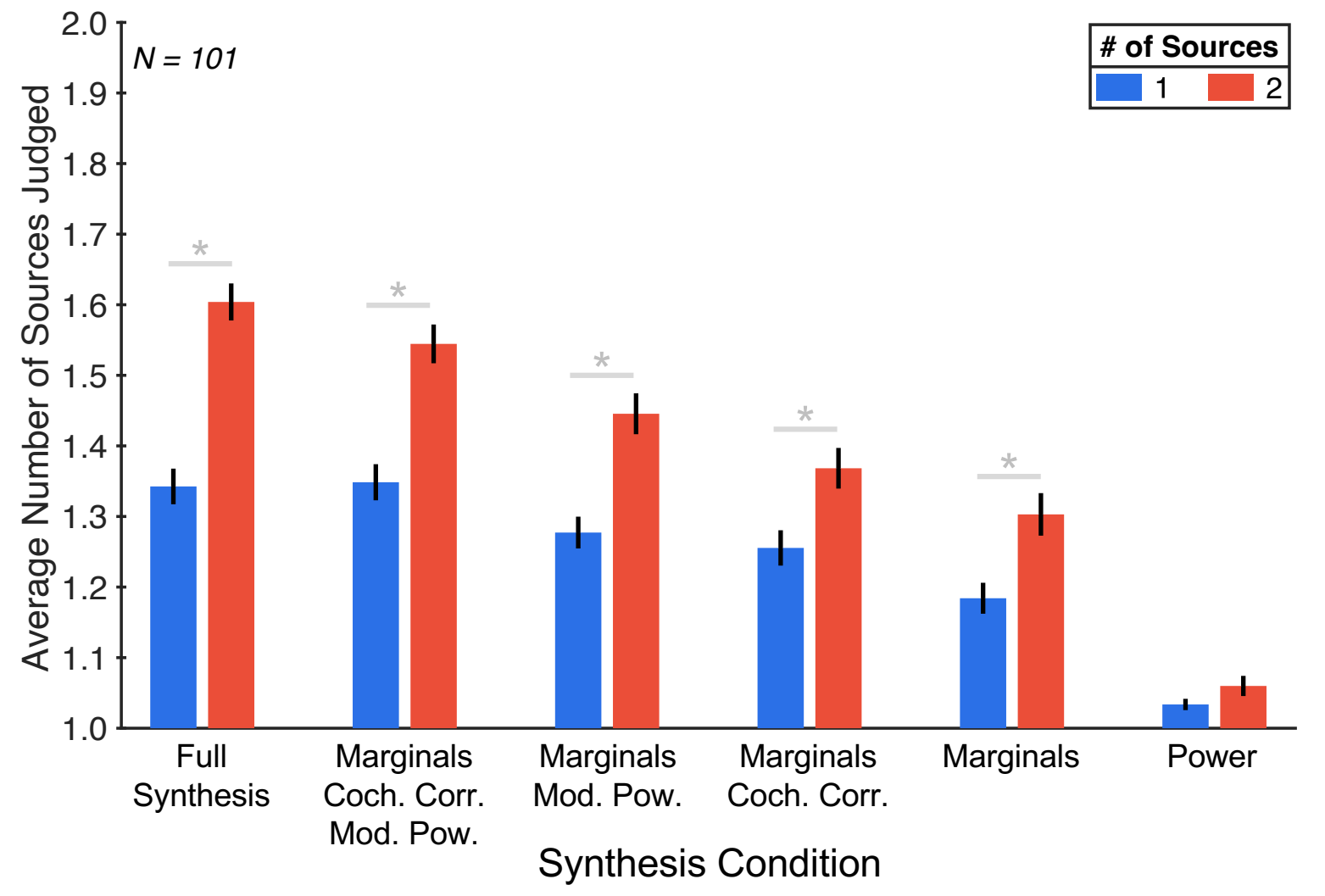
- Listeners can detect concurrent sound textures
- Suggests textures contain sufficient statistical regularities to enable inference of number of sources
- Judgments for individual scenes reveal consistent patterns of errors
- Some combinations of textures tended to be mistakenly heard as a single stream

Is texture streaming dependent on higher-order statistics?

- Synthesized textures by matching different subsets of statistics

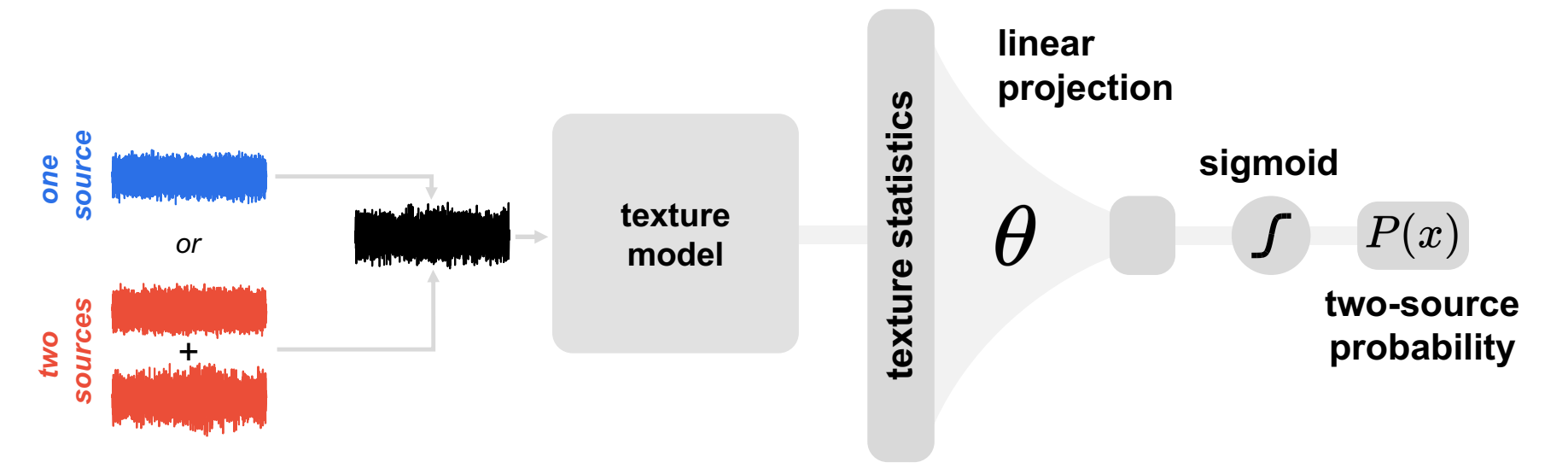


McDermott & Simoncelli (2011) *Neuron*; McWalter & McDermott (2018) *Curr. Biol.*

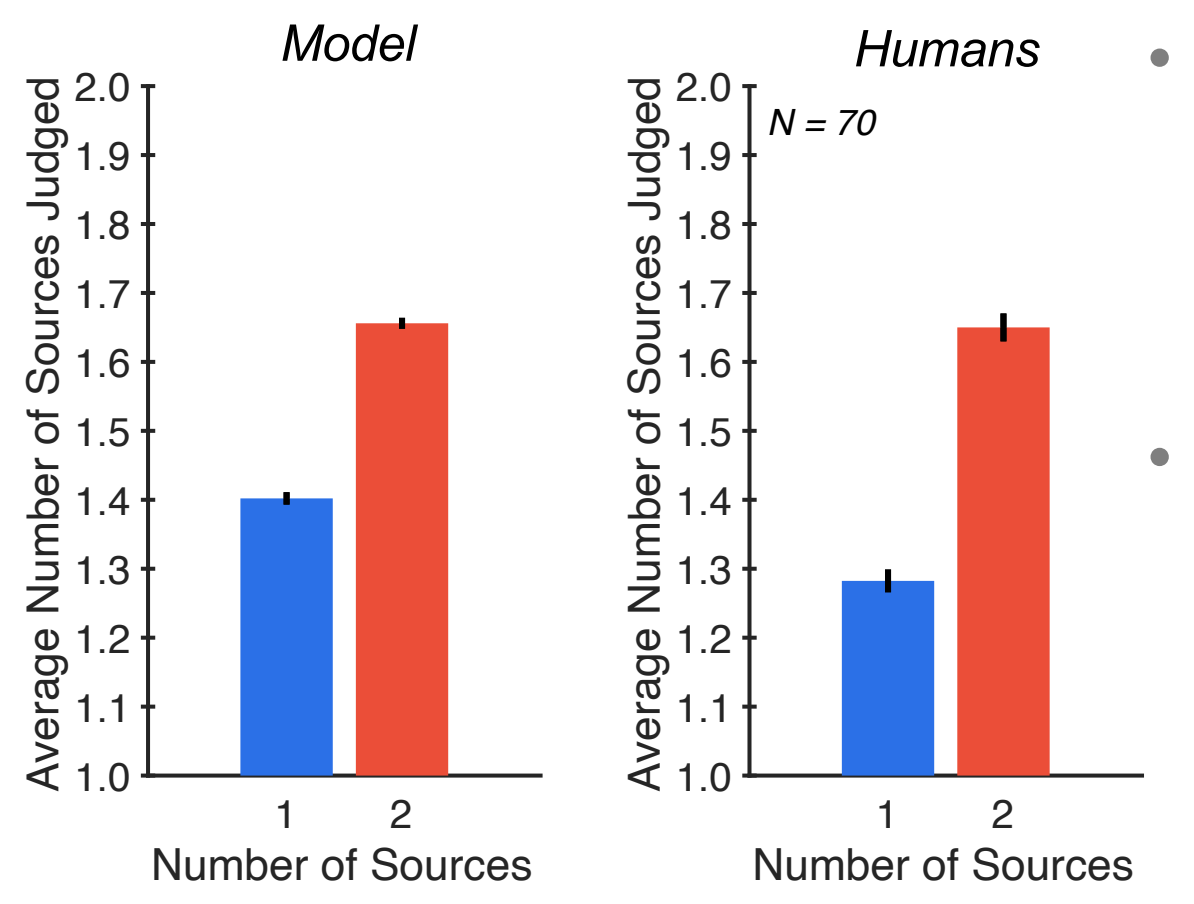


- Synthetic textures defined by subsets of statistics produced weaker streaming than textures in which all statistics in the texture model are matched
- Suggests streaming depends critically on higher-order statistical structure

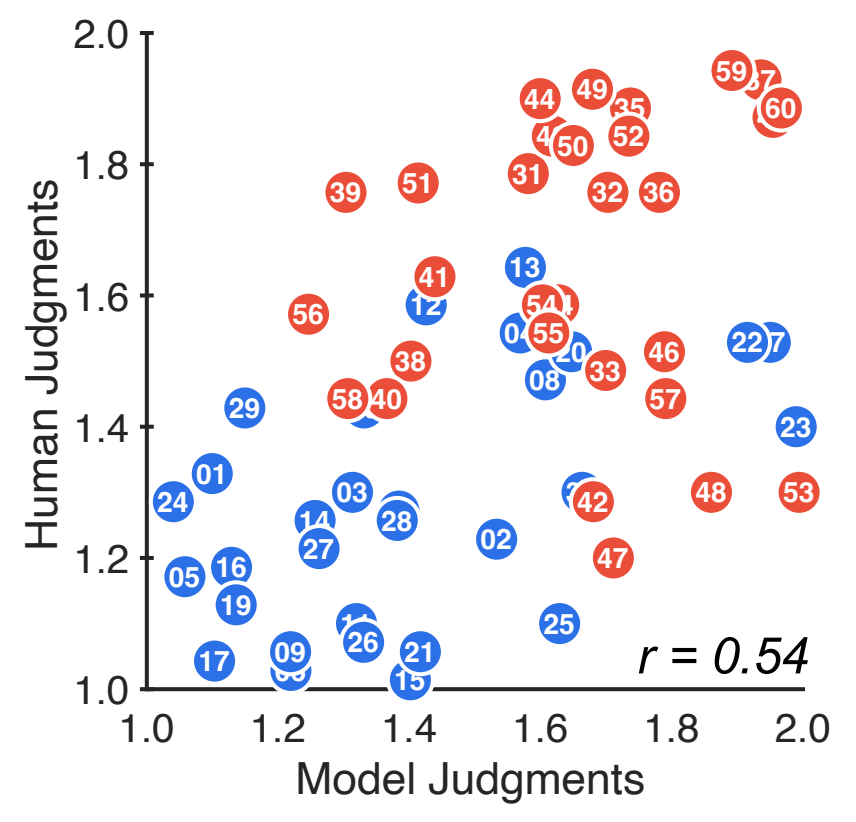
Can an observer model using texture statistics judge the number of sources?



- Trained logistic regression model to discriminate between one & two textures from statistics of stimulus

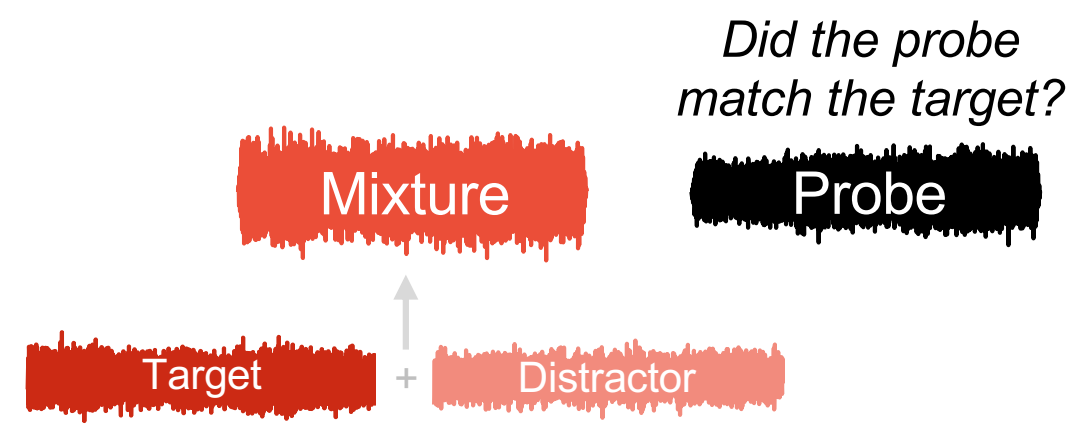


- Model achieved approximately human level performance but could not fully account for judgments on individual stimuli
- Suggests human judgments may be based on alternative computation, e.g. inference in a generative model with a prior over texture statistics

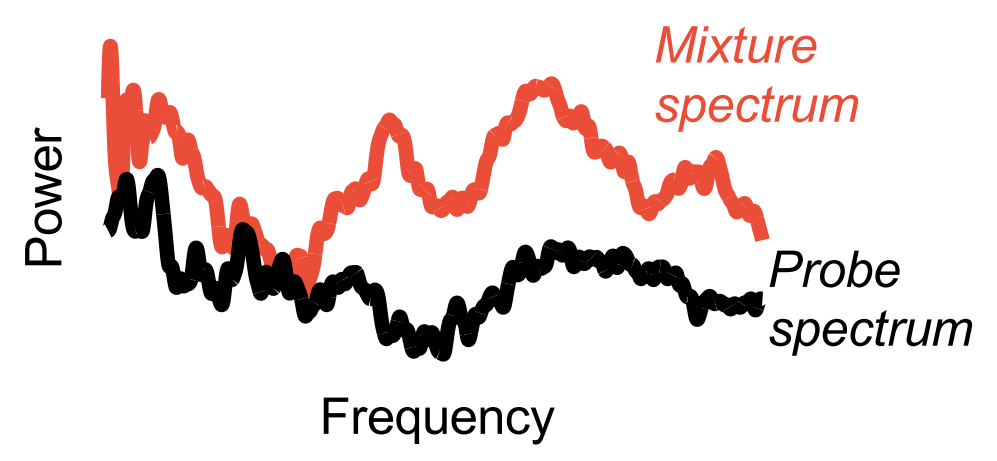


Objective task

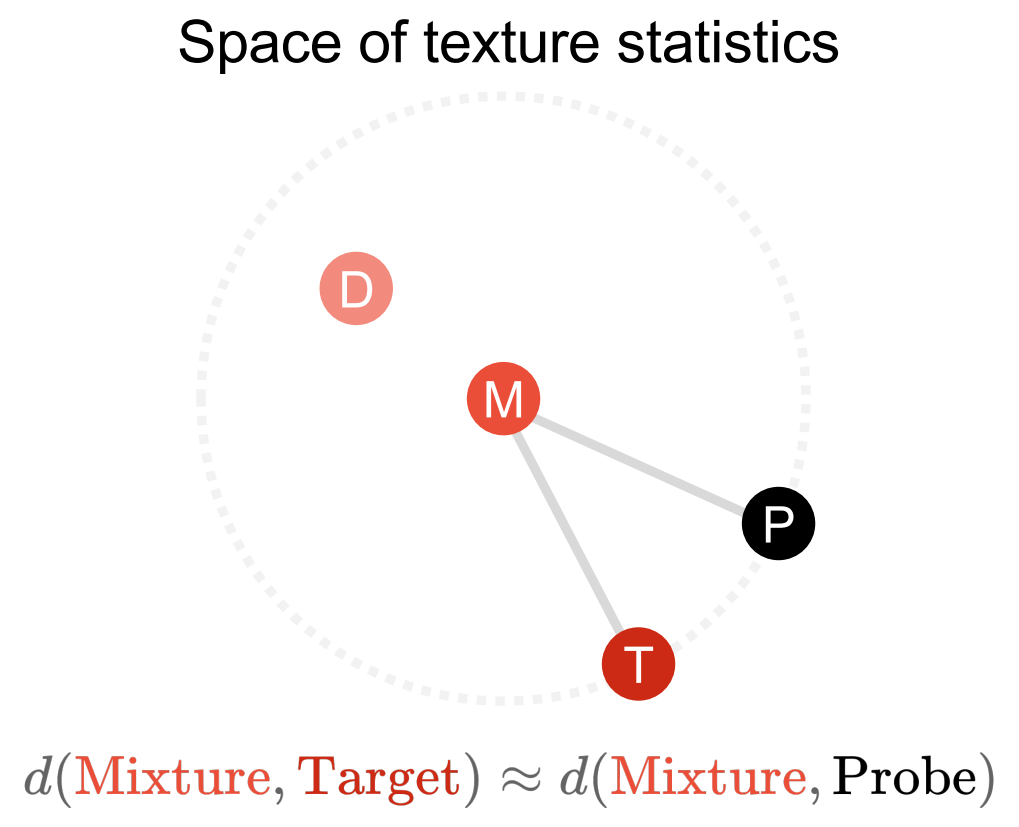
- Developed objective task to measure whether listeners can estimate statistics of a texture from a mixture



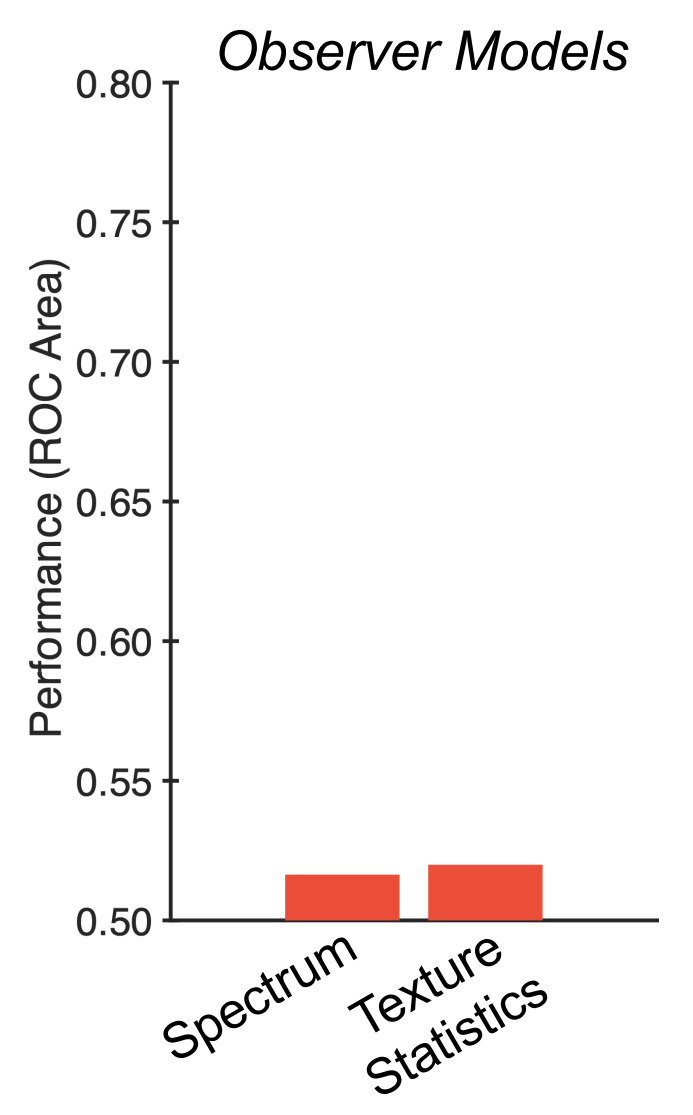
- Incorrect probes constrained:
  1. *To have spectra physically consistent with mixture*



  2. *To be approximately same distance from mixture as target in space of statistics*



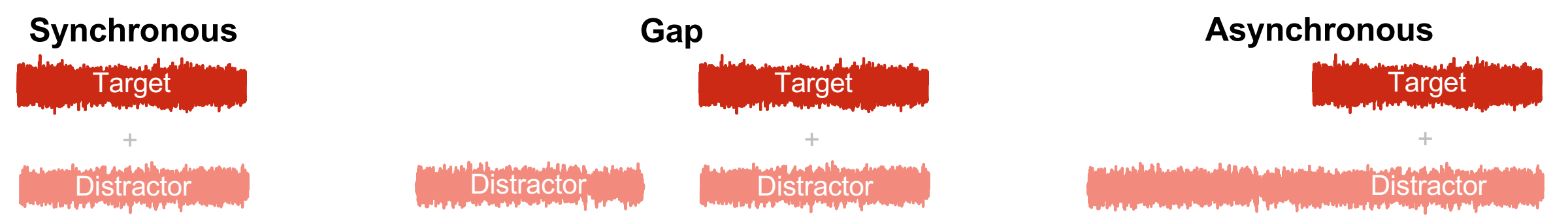
$d(\text{Mixture}, \text{Target}) \approx d(\text{Mixture}, \text{Probe})$



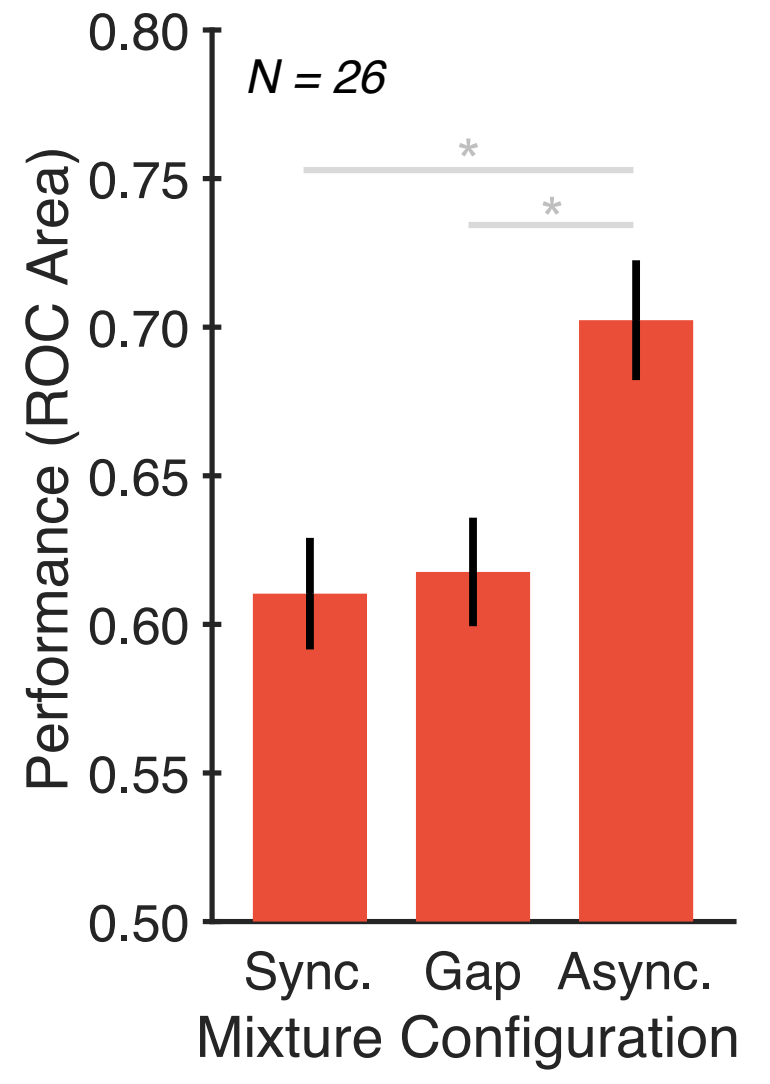
- Observer models comparing either spectrum or texture statistics of mixture to that of the probe performed at chance
- Suggests task could not be performed without successfully "hearing out" the target texture

Can listeners infer the statistics of textures in a mixture?

- Tested three mixture configurations:

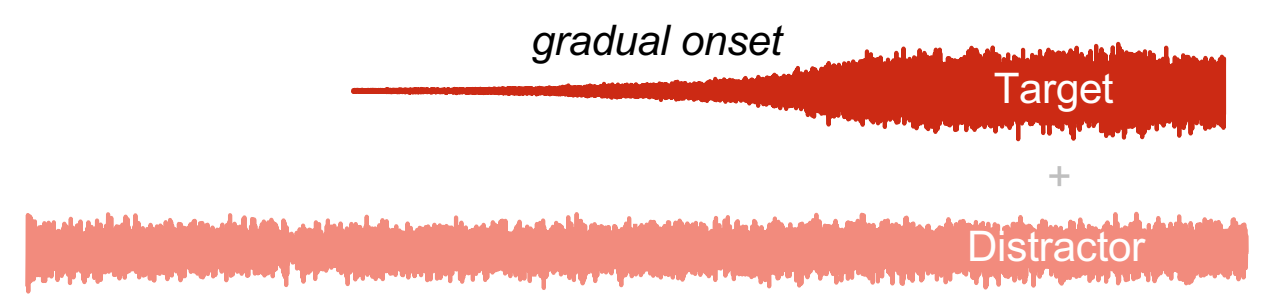


- Listeners had some ability to infer statistics of target but were significantly aided by asynchrony
- Suggests auditory system uses "glimpse" of distractor to estimate target
- Improvement relative to gap configuration suggests the underlying process is perceptual rather than cognitive

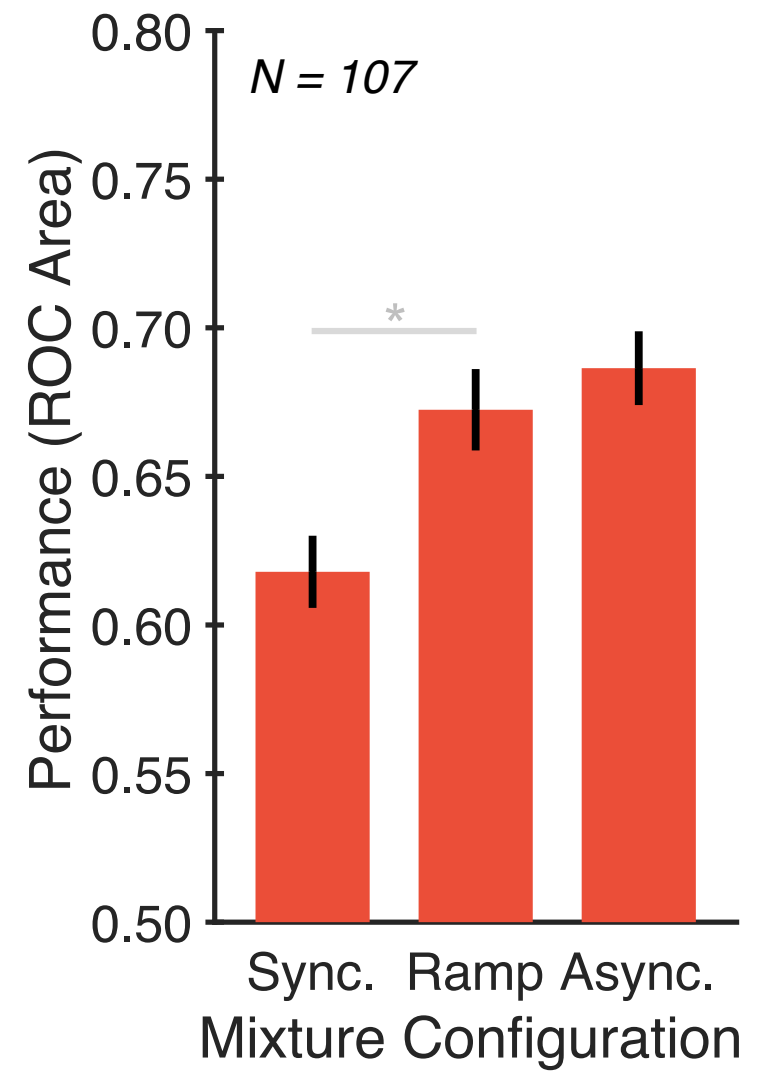


Are abrupt onsets necessary for an asynchrony benefit?

- Introduced target with gradual onset over a 3s interval
- Target amplitude increased linearly at rate of 10 dB/second

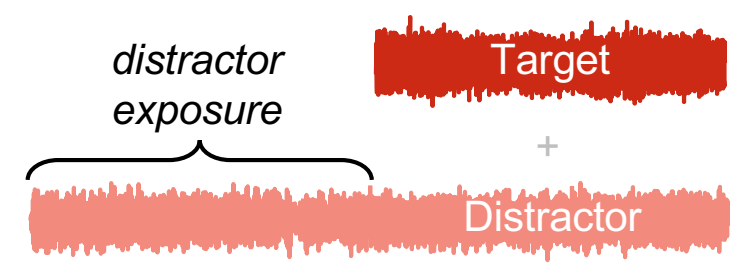


- Asynchrony benefit persisted for gradual onsets
- Suggests broad ecological relevance (given likely prevalence of slow amplitude modulations in everyday conditions)

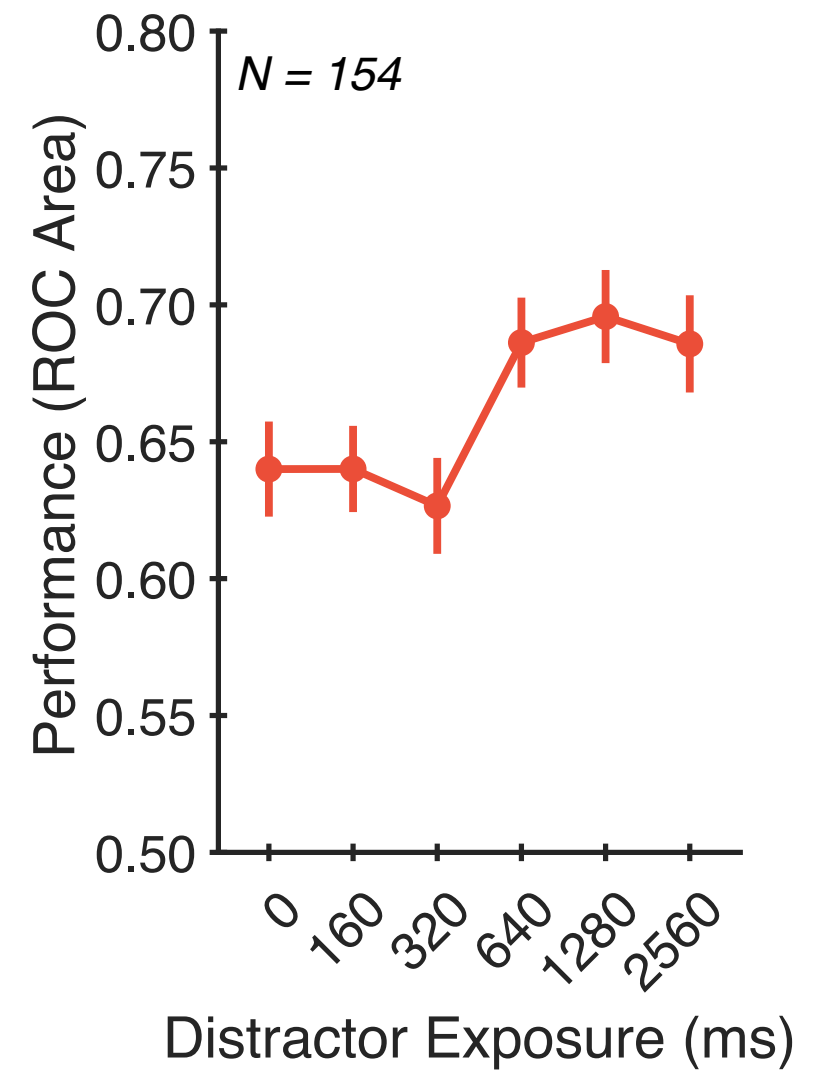


How much exposure is sufficient for an asynchrony benefit?

- Texture properties can be better estimated with longer durations
- If estimation of distractor aids estimation of target, then performance should improve with distractor exposure

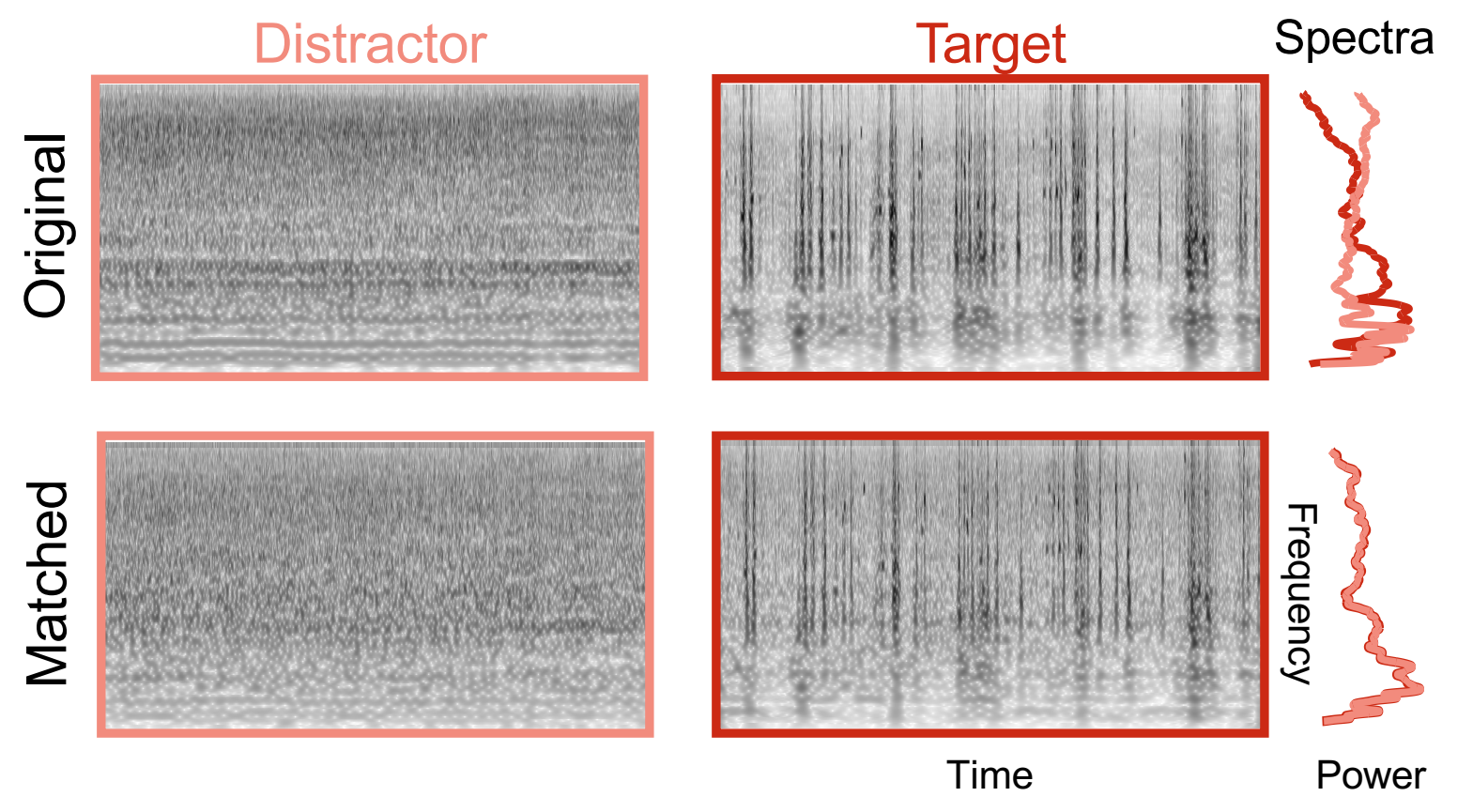


- Manipulated duration of exposure to distractor prior to target onset
- ~1 second of pre-onset exposure was sufficient for an asynchrony benefit
- Suggests a better estimate of one source aids estimation of other source

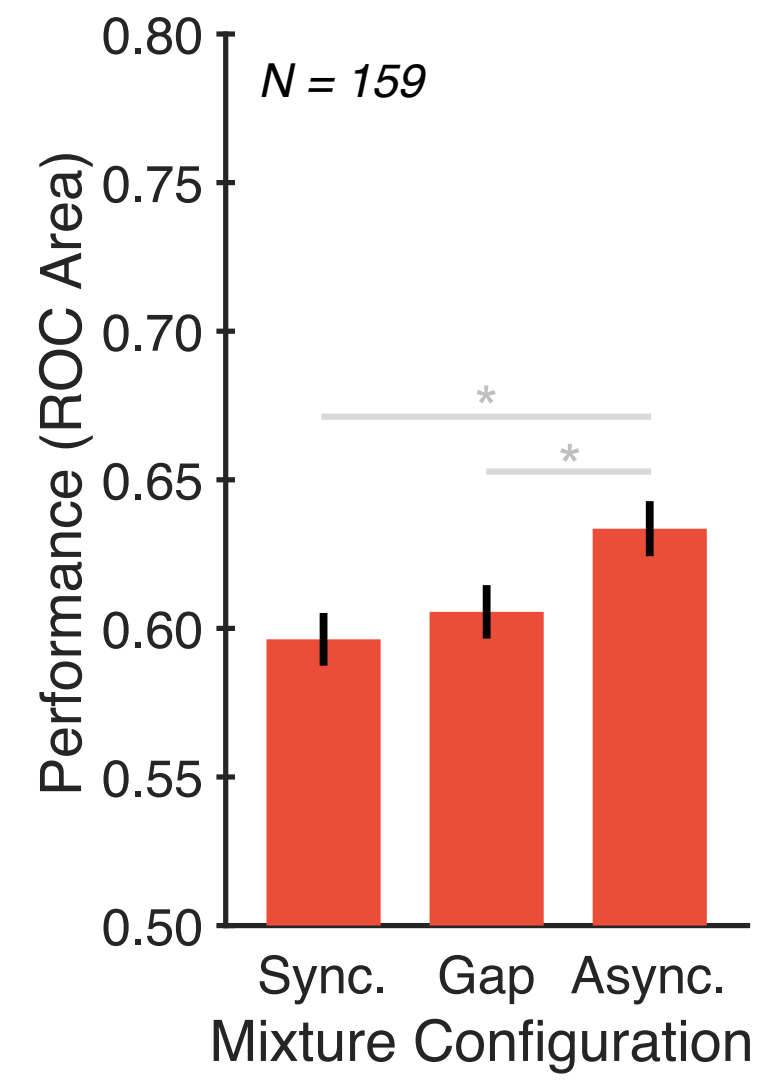


"Old + new" with higher-order statistical properties?

- Matched the spectra of the target, distractor and probes so task could only be performed by estimating higher-order statistics



- Asynchrony benefit persisted, suggesting an "old + new" explanation but with higher-order statistics



Conclusions

- Human listeners could judge the number of sound texture sources in an auditory scene
- Judgments depended critically on higher-order statistical structure of the source textures
- A discriminative model achieved human-level performance but deviated from human judgments on individual stimuli
- Listeners could also estimate statistics of texture sources from a mixture but were aided by onset asynchrony
- Asynchrony benefit persisted for gradual onsets
- Asynchrony benefit increased up to a second of pre-onset exposure to the other texture
- Asynchrony benefit was also evident for mixtures of textures that differed only in higher-order statistics, suggesting an "old + new" explanation with higher-order statistics
- **Ability to infer texture sources appears to leverage principles proposed in classical work with simple synthetic stimuli, but applied to higher-order statistical properties**

Future directions

- Understand what limits performance and whether the auditory system is making sub-optimal judgments
- Development of new models capable of accounting for and explaining these diverse perceptual phenomena

Supported by the NSF Graduate Research Fellowship under Grant No. 1122374 and NSF Grant BCS-2240406